

RESEARCH ARTICLE

Offset Gradient Descent Bit-Flipping Decoding for Irregular Parity-Check Codes

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ABSTRACT A key challenge for decoding irregular parity-check codes under the gradient descent bit-flipping (GDBF) algorithm is that the irregularity in column degrees reduces the accuracy of the reliability measurements obtained through a flipping function (FF). To address the challenge, we introduce a novel approach based on the gradient descent bit-flipping with momentum (GDBF-w/M) algorithm. We first propose an offset GDBF-w/M (OGDBF-w/M) algorithm for decoding column-irregular codes over the additive white Gaussian noise channel, where the FF includes an offset term depending on column degrees. We then propose a diversity-based offset GDBF-w/M (D-OGDBF-w/M) algorithm which employs multiple OGDBF-w/M decoders with distinct offset values. Furthermore, we extend the proposed algorithms to the binary symmetric channel, although they are initially intended for the additive white Gaussian noise channel. Numerical results demonstrate that they achieve significant improvements in decoding performance for irregular parity-check codes, particularly for column-irregular codes.

INDEX TERMS BCH codes, bit-flipping algorithm, irregular parity-check codes, iterative decoding algorithm, LDPC codes.

I. INTRODUCTION

Low-density parity-check (LDPC) codes [1], renowned for their excellent error-correcting capability, are channel codes adopted in various standards including fifth-generation (5G) wireless communications, Wi-Fi, and DVB-S2. When decoded by message-passing (MP) algorithms, they exhibit capacity-approaching performance over various channels such as the additive white Gaussian noise channel (AWGNC) and the Rayleigh fading channel [2], [3], [4]. Herein, MP algorithms generally refer to iterative decoding schemes that exchange and update the messages of variable nodes (VNs) and check nodes (CNs) along the edges of a Tanner graph representing the parity-check matrix (PCM) of an LDPC code. Among them, the sum-product (SP) algorithm as a prominent example demonstrates suboptimal decoding performance for LDPC codes. The min-sum (MS) algorithm is a simplified version of the SP algorithm and has a slight performance degradation. This degradation may be reduced considerably by the offset min-sum (OMS) and

normalized min-sum (NMS) algorithms which introduce an offset and a normalization factor, respectively, to the CN message update of the MS algorithm. In most cases, the computational complexity of an MP algorithm is heavily dependent on the number of quantization bits for the messages. While nonuniform quantization can improve decoding performance [5], [6], it also significantly increases implementation complexity, which motivates the search for simpler decoding schemes.

As a low-complexity alternative, bit-flipping (BF)-based algorithms represent a family of decoders that avoid the need for quantization. These algorithms operate by passing single-bit messages along the edges and flip a set of estimated bits under certain conditions. While MP algorithms update the messages on an edge basis, BF-based algorithms deal with one-bit messages on a node-basis. Despite the significant reduction in computational complexity, BF-based algorithms typically exhibit inferior decoding performance relative to MP algorithms.

The gradient descent BF (GDBF) algorithm [7] formulates the decoding problem as an optimization problem and employs the gradient descent (GD) method to define

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a flipping function (FF). It shows better decoding performance than the conventional BF-based algorithms, but fails to achieve theoretically optimal performance due to the inherent nonlinearity of the problem. This behavior is similar to the situation where the GD method does not guarantee optimal solutions to nonconvex optimization problems. In fact, the decoder under the GDBF algorithm may be trapped in certain states, leading to a decoding failure despite a sufficient number of decoding iterations. To address such a limitation, the probabilistic GDBF (PGDBF) and noisy GDBF (NGDBF) algorithms incorporate randomness into the GDBF algorithm. In the PGDBF algorithm, a group of bits based on the FF is selected, and each bit in the group is flipped with a specific probability [8]. On the other hand, the NGDBF algorithm introduces randomness by adding Gaussian noise to the FF [9].

Another approach to overcome the limit of the GDBF algorithm is to utilize optimization strategies such as variants of tabu search or the momentum method. The tabu-list aided PGDBF (TPGDBF) algorithm combines tabu search with the PGDBF algorithm so that the bits flipped in the current iteration cannot be flipped again in the next iteration [10]. The tabu-list random-penalty GDBF (TRGDBF) algorithm applies a random penalty term to the FF of the GDBF algorithm in conjunction with tabu search [11]. The GDBF-w/M and PGDBF-w/M algorithms incorporate the momentum method into the GDBF and PGDBF algorithms, respectively, by introducing a momentum term into their FFs [12]. The momentum term is a pre-defined penalty applied to the currently flipped bits for a specific number of subsequent iterations. While the deterministic nature of the GDBF-w/M algorithm is advantageous for hardware implementation as it avoids random number generators, it may suffer from an error-floor phenomenon. Some related works have proposed methods to overcome this limitation [13], [14]. Furthermore, the adaptive diversity GDBF (AD-GDBF) algorithm extends the GDBF-w/M algorithm by leveraging decoder diversity [15].

BF-based algorithms work well for column-regular codes whose VNs have a uniform degree, but not for column-irregular codes. Several attempts have been made to mitigate the performance degradation caused by the irregularity in column degrees. In [16], for the binary symmetric channel (BSC), an adaptive probabilistic flipping scheme is utilized, and the VNs of a column-irregular code are classified into two sets based on their degrees, with each set then assigned a distinct weight of the FF. Similarly, in the AD-GDBF algorithm [15], the weights of the FF are optimized in accordance with the column degrees. A generalized AD-GDBF (gAD-GDBF) algorithm for the AWGNC was recently proposed in [17] by introducing an extra reliability measure for each VN. However, these algorithms are not sufficient to fully address the imbalance in which the mean and variance of the FF vary with column degrees. Due to this phenomenon, their decoding

performance for codes with a wide range of column degrees is far from optimal.

To overcome the limitation of the existing algorithms for column-irregular codes, we propose a novel FF that incorporates an offset term based on column degrees, in addition to assigning the weights of the FF according to column degrees. The offset adjusts the mean of the FF, thereby enabling a more effective solution to the imbalance issue associated with the irregularity of column degrees. Furthermore, we present FFs tailored for both the AWGNC and the BSC to ensure their applicability across different channel conditions. The contributions of this paper are summarized as follows.

- We propose an offset GDBF-w/M (OGDBF-w/M) algorithm to enhance decoding performance for column-irregular codes. Existing compensation methods are insufficient to correct the degree-dependent bias in the mean of the FF. Therefore, our proposed OGDBF-w/M algorithm introduces an offset to provide a direct and stable mechanism for adjusting this mean.
- We propose a diversity-based offset GDBF-w/M (D-OGDBF-w/M) algorithm which employs multiple OGDBF-w/M decoders with distinct parameters. The decoder diversity technique provides an additional performance gain.
- We extend the proposed algorithms for the AWGNC to those for the BSC. Our analysis of the extension reveals that while most components of the FF transform antisymmetrically between the two channel models, the offset term behaves symmetrically.

The rest of this paper is organized as follows. In Section II, we describe our system model and review the GDBF algorithm and its variants. The proposed OGDBF-w/M and D-OGDBF-w/M algorithms are presented in Section III. They are then extended to the BSC in Section IV. Numerical results on their decoding performance for irregular parity-check codes are provided in Section V. Finally, concluding remarks are given in Section VI.

Notation: Upper-case and lower-case boldface letters denote matrices and row vectors, respectively. $\mathbb{E}[\cdot]$ and $\mathbb{V}[\cdot]$ stand for the expectation operator and the statistical variance, respectively. For a positive integer A , we denote the set $\{1, 2, \dots, A\}$ by $[A]$. The operator $\oplus$ represents modulo-2 addition for scalar operands and component-wise modulo-2 addition for vector operands.

II. PRELIMINARIES

A. SYSTEM MODEL

Let $\mathbf{H} = (h_{m,n})$ be an $M \times N$ matrix over the binary field $\{0, 1\}$. Then the (N, K) code $\mathcal{C}$ defined by $\mathcal{C} = \{\mathbf{c} \in \{0, 1\}^N \mid \mathbf{H}\mathbf{c}^t = \mathbf{0}\}$ is called a parity-check code or a linear block code, where N is the code length, K is the dimension, and $R = K/N \geq 1 - M/N$ is the code rate. The superscript t stands for the transpose, and every vector is assumed to be a row vector throughout the paper. The matrix $\mathbf{H}$ is referred to as a PCM of $\mathcal{C}$. It is natural

to describe the code $\mathcal{C}$ by the Tanner graph corresponding to $\mathbf{H}$, which is a bipartite graph whose vertices are partitioned into CNs and VNs. For $m \in [M]$, we denote the index set of VNs connected to CN m by $\mathcal{N}(m)$, that is, $\mathcal{N}(m) = \{j \in [N] \mid h_{m,j} = 1\}$. The degree of CN m , denoted by $d(m)$, is defined as the size of $\mathcal{N}(m)$. Similarly, for $n \in [N]$, we denote the index set of CNs connected to VN n by $\mathcal{M}(n)$, that is, $\mathcal{M}(n) = \{i \in [M] \mid h_{i,n} = 1\}$. The degree of VN n is given by the size of $\mathcal{M}(n)$, which is denoted by $d(n)$. When $d(m)$ and $d(n)$ are constant for any $m \in [M]$, $n \in [N]$, the code is said to be regular. Otherwise, it is said to be irregular. In particular, the code is called row-regular if $d(m)$ is constant for any $m \in [M]$. Similarly, the code is referred to as column-regular if $d(n)$ is constant for any $n \in [N]$. Otherwise, it is column-irregular. In this paper, we focus on column-irregular codes.

For the AWGNC, we employ bipolar (+1 or -1) representation. In other words, each component c_n of a codeword $\mathbf{c} \in \mathcal{C}$ is modulated through binary phase-shift keying (BPSK), mapped to $1 - 2c_n$, and transmitted over the AWGNC. Then, the received vector $\mathbf{y} = (y_1, y_2, \dots, y_N) \in \mathbb{R}^N$ is modeled by $\mathbf{y} = (\mathbf{1} - 2\mathbf{c}) + \mathbf{z}$ where $\mathbf{1}$ is the all-one vector of length N , and each component of $\mathbf{z}$ is an independent and identically distributed (i.i.d.) Gaussian random variable with mean zero and variance σ^2 . In order to decode a bipolar codeword $\mathbf{1} - 2\mathbf{c}$ from $\mathbf{y}$, an iterative algorithm may be employed. The estimate at the ℓ -th iteration is denoted by $\mathbf{x}^{(\ell)} = (x_1^{(\ell)}, x_2^{(\ell)}, \dots, x_N^{(\ell)}) \in \{+1, -1\}^N$, where $x_n^{(\ell)}$ corresponds to the n -th VN for $n \in [N]$. At the beginning of the decoding process, $x_n^{(0)}$ is set to $\text{sign}(y_n) \in \{+1, -1\}$ for $n \in [N]$, where $\text{sign}(\cdot)$ is the sign function. The maximum number of iterations, denoted by $\ell_{\max}$, is conventionally prescribed.

In the setting of the BSC, we employ unipolar (0 or 1) representation. When a codeword $\mathbf{c}$ is transmitted over the BSC, the received vector $\mathbf{r} = (r_1, r_2, \dots, r_N) \in \{0, 1\}^N$ is given by $\mathbf{r} = \mathbf{c} \oplus \mathbf{e}$, where $\mathbf{e} = (e_1, e_2, \dots, e_N) \in \{0, 1\}^N$ is an error vector. Note that $\Pr(e_n = 1)$ is the crossover probability of the BSC, and $\Pr(e_n = 0) = 1 - \Pr(e_n = 1)$. Under an iterative decoding algorithm, the estimate of $\mathbf{c}$ at the ℓ -th iteration is given by $\mathbf{x}^{(\ell)} = (x_1^{(\ell)}, x_2^{(\ell)}, \dots, x_N^{(\ell)}) \in \{0, 1\}^N$, where $x_n^{(\ell)}$ corresponds to the n -th VN for $n \in [N]$. Similarly, $x_n^{(0)}$ is initialized to $r_n \in \{0, 1\}$ for all n .

B. GDBF ALGORITHM AND ITS VARIANTS

The GDBF algorithm [7], originally for the AWGNC, is premised on interpreting the decoding process as a discrete optimization problem. A generalized expression of the problem is given in [12] by

$$\underset{(x_1, x_2, \dots, x_N) \in \{\pm 1\}^N}{\text{maximize}} \quad \alpha \sum_{n=1}^N x_n y_n + \sum_{m=1}^M \prod_{j \in \mathcal{N}(m)} x_j,$$

where α is a positive constant. The first term of the objective function is a correlation term based on the maximum

likelihood criterion, and the second term is a penalty term derived from parity-check constraints.

As a strategy to solve the problem, the FF of the GDBF algorithm is defined via the GD method. The FF measures the reliability for VN n at the ℓ -th iteration and is expressed by

$$\Delta_n^{(\ell)} = \alpha x_n^{(\ell)} y_n + \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)}, \quad (1)$$

where $s_m^{(\ell)} \in \{+1, -1\}$ is the bipolar syndrome component corresponding to CN m for $m \in [M]$ at the ℓ -th iteration, that is, $s_m^{(\ell)} = \prod_{j \in \mathcal{N}(m)} x_j^{(\ell)}$. The smaller $\Delta_n^{(\ell)}$, the less reliable VN n . In order to select a group of bits to be flipped, we define $\Delta_{\min}^{(\ell)} = \min_{n \in [N]} \Delta_n^{(\ell)}$ and $\mathcal{F}^{(\ell)} = \{i \in [N] \mid \Delta_i^{(\ell)} \leq \Delta_{\min}^{(\ell)} + \epsilon\}$, where ϵ is a nonnegative flipping margin. At the ℓ -th iteration, $x_n^{(\ell)}$ is flipped to $-x_n^{(\ell)}$ if $n \in \mathcal{F}^{(\ell)}$.

The repeated flipping issue in the GDBF algorithm is a primary cause of decoding failure. When trapped in a local optimum, the GDBF algorithm may fail to escape and certain bits flip repeatedly because its FF is solely based on current iteration information. To address the issue, the GDBF-w/M algorithm [12] utilizes an FF that incorporates a momentum term γ_n into (1). The momentum term helps to mitigate re-flipping of a flipped bit by increasing its FF during a specific number of iterations. The FF of the GDBF-w/M algorithm is calculated as

$$\Delta_n^{(\ell)} = \alpha x_n^{(\ell)} y_n + \beta \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)} + \gamma_n, \quad (2)$$

where β is a positive constant, l_n is the number of iterations from the last flip at VN n , and γ_i is the i -th component of a predetermined momentum vector $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \dots, \gamma_L)$ of length L . Typically, $\gamma_1 \geq \gamma_2 \geq \dots \geq \gamma_L \geq 0$ and the convention $\gamma_i = 0$ if $i > L$ are assumed. At the initial iteration, l_n is set to $L + 1$ for all $n \in [N]$. Note that the FF in (2) is a generalized form of the original FF in [12], where $\beta = 1.0$ yields the original FF.

As shown in [15], the FF of the GDBF-w/M algorithm over the BSC is modified as

$$\check{\Delta}_n^{(\ell)} = \alpha (x_n^{(\ell)} \oplus r_n) + \beta \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)} - \gamma_n, \quad (3)$$

where $\check{s}_m^{(\ell)} = \bigoplus_{j \in \mathcal{N}(m)} x_j^{(\ell)} \in \{0, 1\}$ is the unipolar syndrome component corresponding to CN m for $m \in [M]$ at the ℓ -th iteration. Unlike the AWGNC, the larger $\check{\Delta}_n^{(\ell)}$, the less reliable VN n . Consequently, $x_n^{(\ell)}$ is flipped into $x_n^{(\ell)} \oplus 1$ at the ℓ -th iteration if $n \in \check{\mathcal{F}}^{(\ell)} \triangleq \{i \in [N] \mid \check{\Delta}_i^{(\ell)} \geq \check{\Delta}_{\max}^{(\ell)} - \epsilon\}$, where ϵ is a nonnegative constant and $\check{\Delta}_{\max}^{(\ell)} = \max_{n \in [N]} \check{\Delta}_n^{(\ell)}$. For the BSC, α and β are typically selected as positive integers, and ϵ is set to 0. Note that the momentum term $-\gamma_n$ in (3) plays a role in decreasing the value of the FF. Without it, the algorithm is reduced to the GDBF algorithm over the BSC.

The GDBF and GDBF-w/M algorithms may be applied to irregular codes, but they do not show good decoding performance. To compensate for irregularity, the gAD-GDBF

algorithm [17] employs a decoder diversity scheme with a total of T decoders and defines the FF for the t -th decoder as

$$\Delta_n^{(\ell)} = \alpha_{n,t} x_n^{(\ell)} y_n + \beta_{n,t} \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)} + \gamma_{n,t}, \quad (4)$$

where the decoder index $t \in [T]$ is determined by the iteration index ℓ . The weights $\alpha_{n,t}$ and $\beta_{n,t}$ are positive constants for VN n in Decoder t and they are typically set to be identical for the VNs of the same degree, i.e., $\alpha_{n_1,t} = \alpha_{n_2,t}$ and $\beta_{n_1,t} = \beta_{n_2,t}$ if $d(n_1) = d(n_2)$. The vector $\boldsymbol{\gamma}_t = (\gamma_{1,t}, \gamma_{2,t}, \dots, \gamma_{L,t})$ is a predetermined momentum vector for Decoder t , and $\gamma_{i,t}$ is the i -th component of $\boldsymbol{\gamma}_t$. The conditions $\gamma_{1,t} \geq \gamma_{2,t} \geq \dots \geq \gamma_{L,t} \geq 0$ and the convention $\gamma_{i,t} = 0$ when $i > L$ are typically adopted for all t . Additionally, the gAD-GDBF algorithm fundamentally alters the decision mechanism. Instead of flipping bits, it updates a variable potential for each VN and then determines the new bit value from the sign of that potential in each iteration. This two-step process increases the computational complexity.

III. OGDBF-w/M AND D-OGDBF-w/M ALGORITHMS

Although the GDBF and GDBF-w/M algorithms can be employed for decoding parity-check codes regardless of the code structure, their decoding performance degrades significantly for column-irregular codes. In this section, we propose an OGDBF-w/M algorithm that leverages an offset term to enhance decoding performance for column-irregular codes. It is based on a newly formulated FF. Its diversity-based variant, named D-OGDBF-w/M, is also proposed.

A. COLUMN-IRREGULARITY IN PRACTICAL CODES

Column-irregularity is a key feature of many high-performance codes used in communication systems. This feature is prominently observed in modern codes such as quasi-cyclic LDPC (QC-LDPC) codes as well as in classical codes like Bose–Chaudhuri–Hocquenghem (BCH) codes.

While the LDPC codes initially proposed by Gallager are regular, it was demonstrated in [18] that irregular LDPC codes might achieve much better error-correction performance. Subsequently, QC-LDPC codes with a dual-diagonal structure [19], which were proposed to lower encoding complexity, have been adopted in the standards of Wi-Fi and 5G. The PCM $\mathbf{H}$ of a QC-LDPC code in the IEEE 802.11n standard is partitioned into $\mathbf{H} = [\mathbf{H}_I \mathbf{H}_P]$ where $\mathbf{H}_I$ and $\mathbf{H}_P$ correspond to the information bits and the parity bits, respectively [20]. Specifically, $\mathbf{H}_P$ is designed to have a dual-diagonal structure. Fig. 1 illustrates an example of $\mathbf{H}_P$ with a dual-diagonal structure for QC-LDPC codes when the lifting size is Z .

As $\mathbf{H}_P$ is irregular, $\mathbf{H}$ becomes irregular. Since $\mathbf{H}_P$ has 2 or 3 as its column degree, $\mathbf{H}_I$ contains columns with relatively higher degrees to achieve excellent error-correction performance. For instance, the column degree distribution of the (1944, 972) 802.11n LDPC code [20] spans a wide range and is given by $\Lambda(z) \triangleq \sum_i \Lambda_i z^i = 891z^2 + 729z^3 +$

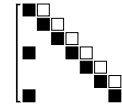


FIGURE 1. An example of $\mathbf{H}_P$ with a dual-diagonal structure. The white box($\square$) and the black box($\blacksquare$) represent the $Z \times Z$ identity matrix and a $Z \times Z$ circulant permutation matrix, respectively.

$81z^4 + 243z^{11}$, where Λ_i is the number of columns with degree i in $\mathbf{H}$. The corresponding row degree distribution is expressed as $\Gamma(z) \triangleq \sum_i \Gamma_i z^i = 810z^7 + 162z^8$, where Γ_i is the number of rows with degree i in $\mathbf{H}$. It is well-known in [3] that the row degrees of well-designed LDPC codes tend to be concentrated around a few specific values.

BCH codes are known as a family of algebraic codes having good error-correction capability [21]. Since they are cyclic, their PCMs can be designed by cyclically shifting a single row. The column degree distribution of the PCM constructed in this manner is highly irregular, but all the rows have an identical degree. For example, the (63, 45) BCH code [22] has $\Lambda(z) = 2z + 6z^2 + 2z^3 + 4z^4 + 7z^5 + 5z^6 + 9z^7 + 6z^8 + 6z^9 + 10z^{10} + 6z^{11}$ and $\Gamma(z) = 18z^{24}$.

B. NECESSITY OF INTRODUCING AN OFFSET

A fundamental question arises as to why the GDBF and GDBF-w/M algorithms do not work well for column-irregular codes. In order to answer this question, we analyze the bipolar syndrome sum $S_n^{(\ell)} \triangleq \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)}$ in (1) and (2). For a tractable analysis, we assume the following:

- 1) The all-zero codeword is transmitted;
- 2) The considered code is row-regular;
- 3) Its PCM is sufficiently sparse so that the syndromes corresponding to the neighboring CNs of a VN are independent;
- 4) The bit error probability is assumed to be identical for all VNs sharing the same column degree at each iteration; and
- 5) The degree distribution of the VNs connected to a CN is identical for all CNs.

The following lemma [1] provides insight into the properties of $s_m^{(\ell)}$.

Lemma 1 ([1], [23, Chap. 5]): Let $b_1, b_2, \dots, b_L$ be independent random variables taking values of 0 or 1 with $\Pr(b_i = 1) = p_i$. Then the probability that $\mathbf{b} = (b_1, b_2, \dots, b_L)$ contains an even number of 1s is

$$\frac{1}{2} + \frac{1}{2} \prod_{i=1}^L (1 - 2p_i), \quad (5)$$

and the probability that $\mathbf{b}$ contains an odd number of 1s is

$$\frac{1}{2} - \frac{1}{2} \prod_{i=1}^L (1 - 2p_i). \quad (6)$$

The unipolar representation is transformed into the bipolar representation via the mapping given by $0 \leftrightarrow +1$ and $1 \leftrightarrow -1$. By this mapping, Assumption 1) is equivalent to the

transmission of the all-one bipolar codeword, and the modulo-2 addition in the unipolar domain is equivalent to the product in the bipolar domain. Therefore, by Lemma 1, the probability that CN m is unsatisfied at the ℓ -th iteration is given by

$$q_m^{(\ell)} \triangleq \Pr(s_m^{(\ell)} = -1) = \frac{1}{2} - \frac{1}{2} \prod_{i \in \mathcal{N}(m)} (1 - 2p_i^{(\ell)}), \quad (7)$$

where the error probability of VN n at iteration ℓ is $p_n^{(\ell)} \triangleq \Pr(x_n^{(\ell)} = -1)$. Conventionally, we are interested in the situation when $p_n^{(0)} < 1/2$. This is because the signs of the received bits are initially inverted when the error probability at the channel is greater than $1/2$. Also, $p_n^{(\ell)} < 1/2$ for $\ell > 0$, because the error probability tends to decrease with decoding iterations. Therefore, we have $q_m^{(\ell)} < 1/2$ for all ℓ . Under the five assumptions, $q_m^{(\ell)}$ is independent of m , i.e., $q_m^{(\ell)} = q^{(\ell)} < 1/2$.

Since $s_m^{(\ell)}$ can be modeled as an i.i.d. random variable with $\Pr(s_m^{(\ell)} = -1) = q^{(\ell)}$ and $\Pr(s_m^{(\ell)} = +1) = 1 - q^{(\ell)}$, the expectation and variance of $S_n^{(\ell)}$ are $\mathbb{E}[S_n^{(\ell)}] = d(n)(1 - 2q^{(\ell)}) > 0$ and $\mathbb{V}[S_n^{(\ell)}] = 4d(n)q^{(\ell)}(1 - q^{(\ell)})$, respectively. Clearly, both $\mathbb{E}[S_n^{(\ell)}]$ and $\mathbb{V}[S_n^{(\ell)}]$ are proportional to $d(n)$. Therefore, the dependency of $S_n^{(\ell)}$ on $d(n)$ should be properly compensated for to ensure that the FF serves as a fair reliability metric.

An intuitive approach to compensate for the degree-dependent nature of $S_n^{(\ell)}$ is to introduce a degree-dependent scaling factor, $\beta_n > 0$, as in the gAD-GDBF algorithm. The expectation and variance of the scaled syndrome sum $\beta_n S_n^{(\ell)}$ are then $\mathbb{E}[\beta_n S_n^{(\ell)}] = \beta_n d(n)(1 - 2q^{(\ell)})$ and $\mathbb{V}[\beta_n S_n^{(\ell)}] = 4\beta_n^2 d(n)q^{(\ell)}(1 - q^{(\ell)})$, respectively. Clearly, the expectation is proportional to β_n while the variance is proportional to β_n^2 . Hence, it is not possible to find the values of β_n which simultaneously equalize both the expectations and the variances for VNs with distinct degrees.

One might consider α in (2) to be degree-dependent, by replacing it with the parameter $\alpha_n > 0$. Let us assume that β_n has been chosen to equalize the variances for VNs with distinct $d(n)$. The remaining task is to adjust α_n so that the expectations for VNs with distinct degrees are equalized. The first term of the FF in (1) and (2) is either $+\alpha_n|y_n|$ or $-\alpha_n|y_n|$ depending on $x_n^{(\ell)}$, while the distribution of y_n is identical for all n . This presents a trade-off: If a large value of α_n is chosen to boost a small expectation (i.e., to provide compensation in the case of $+\alpha_n|y_n|$), a large negative component is simultaneously introduced in the case where the n -th bit flips, leading to further degradation. Choosing a small α_n to mitigate this negative impact is also insufficient, as it would diminish the desired compensatory effect. Due to such a “double-edged sword” effect, it is not a stable compensation method to adjust α_n .

Based on the preceding analysis, it is necessary to introduce an offset δ_n . By jointly applying the degree-dependent parameters β_n and δ_n , it becomes possible to simultaneously equalize both the mean and variance of $\beta_n S_n^{(\ell)} - \delta_n$ for VNs with distinct $d(n)$. This idea motivates us to propose the OGDBF-w/M algorithm in the next subsection. Note that the momentum term is a temporal penalty to prevent a bit from being flipped repeatedly, while the offset is a spatial penalty to compensate for the effect of column-irregularity.

C. OGDBF-w/M DECODING

As remarked in the previous subsection, the proposed OGDBF-w/M algorithm is based on a newly formulated FF that introduces a degree-dependent offset term to compensate for column-irregularity. The specific design of this offset scheme is motivated by the empirical behavior of the bipolar syndrome sum $S_n^{(\ell)}$. The degree-dependent behavior of $S_n^{(\ell)}$ is confirmed experimentally, although a theoretical analysis in the previous subsection relies on the five simple assumptions.

Fig. 2 shows how the empirical average of $S_n^{(\ell)}$ taken over the VNs with column degree $d(n)$ varies with the number of iterations. The (1944, 972) 802.11n LDPC code and the (63, 45) BCH code are tested over the AWGNC using the GDBF-w/M algorithm (parameters are detailed in Section V), when E_b/N_0 is 5.0 dB and 6.0 dB, respectively. As predicted theoretically, it is confirmed empirically that $\mathbb{E}[S_n^{(\ell)}]$ is heavily dependent on $d(n)$. In particular, the numerical results for the (1944, 972) 802.11n LDPC code in Fig. 2(a) align well with our theoretical analysis since the PCM of the code is so sparse that Assumption 3) is almost satisfied. At initial iterations, $p_n^{(\ell)}$ is so high that the differences among the averages for distinct values of $d(n)$ are not particularly apparent. As ℓ increases, the decrease in $p_n^{(\ell)}$ causes $q^{(\ell)}$ to approach zero, which in turn leads the theoretical mean $\mathbb{E}[S_n^{(\ell)}] = d(n)(1 - 2q^{(\ell)})$ to converge toward $d(n)$, as clearly shown in Fig. 2(a). A similar spreading phenomenon is manifestly shown in Fig. 2(b) for the (63, 45) BCH code. Specifically, the averages of $S_n^{(\ell)}$ for distinct values of $d(n)$ remain relatively close when ℓ is small. Although the dense PCM of the (63, 45) BCH code prevents these averages from fully reaching the limit $d(n)$, a spreading phenomenon appears clear as ℓ increases. This observation confirms that the degree-dependent disparity in $S_n^{(\ell)}$ is an apparent phenomenon even when the PCM is not sparse, thereby justifying the use of degree-dependent offsets to compensate for the disparity.

Based on the above observations and the analytical results in the previous subsection, we propose an OGDBF-w/M algorithm with the offset-based FF given by

$$\Delta_n^{(\ell)} = \begin{cases} \alpha x_n^{(\ell)} y_n + \beta \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)} + \gamma_{l_n}, & \text{if } \ell < \ell_{\text{th}} \\ \alpha_n x_n^{(\ell)} y_n + \beta_n \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)} + \gamma_{l_n} - \delta_n, & \text{otherwise,} \end{cases} \quad (8)$$

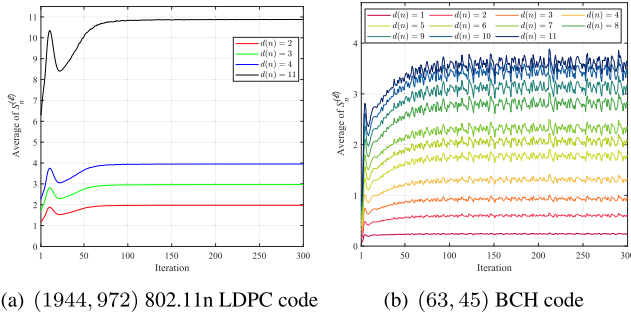


FIGURE 2. Average of $S_n^{(l)}$ taken over the VNs with degree $d(n)$.

where ℓ_{th} is a predetermined iteration threshold, and δ_n is a nonnegative offset for VN n . The FF of the proposed OGDBF-w/M algorithm operates in two phases. During the initial phase ($\ell < \ell_{th}$), the FF of the GDBF-w/M algorithm is used. For iteration $\ell \geq \ell_{th}$, an offset term is applied to the FF based on the gAD-GDBF algorithm [17], which incorporates degree-dependent weights α_n and β_n . The offset δ_n is typically set to be identical for VNs with the same column degree, i.e., $\delta_{n_1} = \delta_{n_2}$ if $d(n_1) = d(n_2)$. Note that (8) is reduced to (2) for a column-regular code. This is because we choose $\alpha_n = \alpha$ and $\beta_n = \beta$ for all n , and we set the offset δ_n normally to zero for any n .

The proposed OGDBF-w/M algorithm may be implemented so that an increase in the number of parameters is minimized, compared to the GDBF-w/M algorithm. For this purpose, we consider a simple parameter sharing approach to α_n and β_n , given by $\alpha_n = \alpha$ and $\beta_n = \beta$ for all n . The same value of the offset δ_n is shared with the VNs of the same column degree, that is, $\delta_{n_1} = \delta_{n_2}$ if $d(n_1) = d(n_2)$. Additionally, δ_n is chosen to be nonnegative and increases monotonically with $d(n)$, i.e., $\delta_{n_1} \leq \delta_{n_2}$ if $d(n_1) \leq d(n_2)$. This is because $\mathbb{E}[S_n^{(l)}]$ is positive and proportional to $d(n)$. Specifically, such monotonicity is expected when α and β are fixed as constants. However, it is not necessarily required when δ_n is jointly optimized with other degree-dependent parameters such as α_n or β_n .

It is noteworthy that the proposed OGDBF-w/M algorithm has the same flipping process as that of the GDBF-w/M algorithm. Furthermore, the OGDBF-w/M algorithm is distinguished from the gAD-GDBF algorithm in the sense that the former does not require additional computation of variable potentials while the latter does. The OGDBF-w/M algorithm is summarized in Algorithm 1.

D. D-OGDBF-w/M DECODING

The parameters α_n , β_n , and δ_n in the OGDBF-w/M algorithm are chosen to be fixed even for $\ell \geq \ell_{th}$. However, this choice of fixed parameters makes the OGDBF-w/M algorithm suboptimal, since $\mathbb{E}[S_n^{(l)}]$ and $\mathbb{V}[S_n^{(l)}]$ depend on $q^{(\ell)}$ as well as $d(n)$. In order to enhance its decoding performance, we propose a D-OGDBF-w/M algorithm employing decoder diversity as in the gAD-GDBF algorithm. Consider T decoders operating with the OGDBF-w/M

Algorithm 1 OGDBF-w/M Algorithm Over the AWGNC

```

1: Initialization:  $\ell = 0$ ,  $x_n^{(0)} = \text{sign}(y_n)$ , and  $l_n = L + 1$ 
   for all  $n \in [N]$ 
2: Calculate  $s_m^{(\ell)}$  for all  $m \in [M]$ ;
3: if  $(s_1^{(\ell)}, s_2^{(\ell)}, \dots, s_M^{(\ell)})$  is all one vector then
4:   stop and output  $(x_1^{(\ell)}, x_2^{(\ell)}, \dots, x_N^{(\ell)})$ ;
5: end if
6: for  $n = 1$  to  $N$  do
7:    $l_n \leftarrow \min(l_n, L) + 1$ ;
8:   Calculate  $\Delta_n^{(\ell)}$  using (8);
9: end for
10: Calculate  $\Delta_{\min}^{(\ell)} = \min_{n \in [N]} \Delta_n^{(\ell)}$ ;
11: for  $n = 1$  to  $N$  do
12:   if  $\Delta_n^{(\ell)} \leq \Delta_{\min}^{(\ell)} + \epsilon$  then
13:      $x_n^{(\ell+1)} \leftarrow -x_n^{(\ell)}$  and  $l_n = 0$ ;
14:   else
15:      $x_n^{(\ell+1)} \leftarrow x_n^{(\ell)}$ ;
16:   end if
17: end for
18: Set  $\ell \leftarrow \ell + 1$ ;
19: if  $\ell < \ell_{\max}$  then
20:   go to line 2;
21: else
22:   stop and output  $(x_1^{(\ell)}, x_2^{(\ell)}, \dots, x_N^{(\ell)})$ ;
23: end if

```

algorithm. For $t \in [T]$, the FF of the n -th VN in the t -th decoder is given by

$$\Delta_n^{(\ell)} = \alpha_{n,t} x_n^{(\ell)} y_n + \beta_{n,t} \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)} + \gamma_{l_n,t} - \delta_{n,t}, \quad (9)$$

where Decoder t has a quadruplet of parameters, $(\alpha_{n,t}, \beta_{n,t}, \gamma_{l_n,t}, \delta_{n,t})$, together with $\ell_{th} = 0$.

The decoders operate sequentially, that is, if a decoder does not produce a valid codeword within a prescribed number of iterations, the next decoder in the sequence begins its decoding procedure. Throughout the paper, the above decoding process is referred to as the D-OGDBF-w/M decoding algorithm. The maximum number of iterations for this algorithm is set to $\sum_{t \in [T]} \ell_{\max,t}$, where $\ell_{\max,t}$ is the maximum number of iterations for the t -th decoder. The decoder index t is determined by the iteration index ℓ and is only involved in selecting the parameters $(\alpha_{n,t}, \beta_{n,t}, \gamma_{l_n,t}, \delta_{n,t})$. The beginning iteration index for the t -th decoder, denoted by $\ell_{\text{begin},t}$, is given by

$$\ell_{\text{begin},t} \triangleq \begin{cases} 0, & \text{if } t = 1 \\ \sum_{i=1}^{t-1} \ell_{\max,i}, & \text{if } 2 \leq t \leq T \end{cases}. \quad (10)$$

IV. EXTENSION TO BSC

The OGDBF-w/M and D-OGDBF-w/M algorithms, described for the AWGNC in the previous section, can be readily extended to the BSC. This section first justifies the need for an offset term in the BSC environment from

two different perspectives. Then the FFs of the OGDBF-w/M and D-OGDBF-w/M algorithms over the BSC are presented.

A. JUSTIFICATION FOR THE OFFSET TERM OVER THE BSC

First, an algebraic transformation naturally yields a degree-dependent term that acts as an intrinsic offset. Adapting the GDBF-w/M algorithm from the AWGNC to the BSC begins with establishing the relationship of the correlation between $x_n^{(\ell)}$ and the channel output. For the AWGNC, the correlation $x_n^{(\ell)} y_n$ is a real value as shown in (2), whereas for the BSC, the correlation $x_n^{(\ell)} \oplus r_n$ is binary. To bridge this, we first quantize $x_n^{(\ell)} y_n$ into a bipolar value. It is mapped to +1 if the correlation is strong ($x_n^{(\ell)} y_n \geq 0$) and to -1 if the correlation is weak ($x_n^{(\ell)} y_n < 0$). In the unipolar domain, strong and weak correlations correspond to $x_n^{(\ell)} \oplus r_n = 0$ and $x_n^{(\ell)} \oplus r_n = 1$, respectively. The relationships between the bipolar terms and their unipolar equivalents are $\text{sign}(x_n^{(\ell)} y_n) \leftrightarrow 1 - 2(x_n^{(\ell)} \oplus r_n)$ and $s_m^{(\ell)} \leftrightarrow 1 - 2\check{s}_m^{(\ell)}$. Thus, minimizing the FF in (2) is transformed into a maximization problem as

$$\begin{aligned} & \min_{n \in [N]} \alpha \text{sign}(x_n^{(\ell)} y_n) + \beta \sum_{m \in \mathcal{M}(n)} s_m^{(\ell)} + \gamma_{l_n} \\ &= \min_{n \in [N]} \alpha (1 - 2(x_n^{(\ell)} \oplus r_n)) + \beta \sum_{m \in \mathcal{M}(n)} (1 - 2\check{s}_m^{(\ell)}) + \gamma_{l_n} \\ &= \min_{n \in [N]} -2\alpha (x_n^{(\ell)} \oplus r_n) - 2\beta \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)} + \beta d(n) + \gamma_{l_n} \\ &= \max_{n \in [N]} \alpha (x_n^{(\ell)} \oplus r_n) + \beta \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)} - \beta d(n)/2 - \gamma_{l_n}/2. \end{aligned}$$

As shown in the derivation above, $-\beta d(n)/2$ should not be ignored for column-irregular codes because it is proportional to $d(n)$. Note that the scaling factor $1/2$ in $\gamma_{l_n}/2$ can be omitted by subsuming it into the pre-optimized γ_{l_n} .

Second, a probabilistic analysis of the unipolar syndrome sum term, $\check{S}_n^{(\ell)} \triangleq \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)}$, also shows that its statistics are proportional to $d(n)$. Thus, a compensation term that accounts for this degree-dependency is required. We consider the same assumptions used in the previous section. By Lemma 1, the probability that CN m is unsatisfied at the ℓ -th iteration is defined as

$$\check{q}_m^{(\ell)} \triangleq \Pr(\check{s}_m^{(\ell)} = 1) = \frac{1}{2} - \frac{1}{2} \prod_{i \in \mathcal{N}(m)} (1 - 2\check{p}_i^{(\ell)}), \quad (11)$$

where $\check{p}_n^{(\ell)} \triangleq \Pr(x_n^{(\ell)} = 1)$ is the error probability of VN n at the ℓ -th iteration and $\check{p}_n^{(\ell)} < 1/2$ for all ℓ by convention. Under the assumptions, $\check{q}_m^{(\ell)}$ is independent of m , (i.e., $\check{q}_m^{(\ell)} = \check{q}^{(\ell)} > 0$ provided that any decoding error exists), and $\check{s}_m^{(\ell)}$ is modeled as an i.i.d Bernoulli random variable with $\Pr(\check{s}_m^{(\ell)} = 1) = \check{q}^{(\ell)}$. Therefore, $\check{S}_n^{(\ell)}$ follows the binomial distribution with parameter $d(n)$ and $\check{q}^{(\ell)}$. The expectation and variance of $\check{S}_n^{(\ell)}$ are $\mathbb{E}[\check{S}_n^{(\ell)}] = d(n)\check{q}^{(\ell)} > 0$ and $\mathbb{V}[\check{S}_n^{(\ell)}] = d(n)\check{q}^{(\ell)}(1 - \check{q}^{(\ell)})$, respectively.

As observed over the AWGNC, this imbalance necessitates the introduction of a nonnegative, subtractive, and degree-dependent offset term.

B. OGDBF-w/M AND D-OGDBF-w/M ALGORITHMS OVER THE BSC

Both the algebraic and probabilistic analyses conclude that a degree-dependent offset is necessary for column-irregular codes. We generalize these effects into a single offset term. Similar to the AWGNC, the FF of OGDBF-w/M algorithm over the BSC is given by

$$\check{\Delta}_n^{(\ell)} = \begin{cases} \alpha (x_n^{(\ell)} \oplus r_n) + \beta \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)} - \gamma_{l_n}, & \text{if } \ell < \ell_{\text{th}} \\ \alpha_n (x_n^{(\ell)} \oplus r_n) + \beta_n \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)} - \gamma_{l_n} - \delta_n, & \text{otherwise,} \end{cases} \quad (12)$$

where δ_n is a nonnegative offset for VN n . The OGDBF-w/M algorithm over the BSC follows the same decoding procedure as that of the GDBF-w/M algorithm over the BSC, except for the FF.

A comparison of the proposed OGDBF-w/M algorithm over the AWGNC and the BSC reveals a notable symmetry in the offset term. As summarized in Table 1, most components of the FF exhibit an antisymmetric relationship. The minimization problem for the AWGNC corresponds to a maximization problem for the BSC, and consequently, the flipping margins $+\epsilon$ and $-\epsilon$ are antisymmetric. Similarly, the correlation and syndrome terms, as well as the momentum term, have corresponding antisymmetric forms. The offset term, however, is symmetric. It is subtracted ($-\delta_n$) in both the AWGNC and the BSC. Thus, the offset term is a fundamental and necessary component that must be independently introduced in both environments to compensate for the structural imbalance caused by column-irregularity.

TABLE 1. Comparison of the OGDBF-w/M algorithm over the AWGNC and BSC.

	AWGNC (bipolar)	BSC (unipolar)
Flipping condition	$\Delta_n^{(\ell)} \leq \Delta_{\min}^{(\ell)} + \epsilon$	$\check{\Delta}_n^{(\ell)} \geq \check{\Delta}_{\max}^{(\ell)} - \epsilon$
Correlation is strong (weak)	$x_n^{(\ell)} y_n \geq 0$ $(x_n^{(\ell)} y_n < 0)$	$x_n^{(\ell)} \oplus r_n = 0$ $(x_n^{(\ell)} \oplus r_n = 1)$
CN m is satisfied (unsatisfied)	$s_m^{(\ell)} = +1$ $(s_m^{(\ell)} = -1)$	$\check{s}_m^{(\ell)} = 0$ $(\check{s}_m^{(\ell)} = 1)$
Momentum term	$+\gamma_{l_n}$	$-\gamma_{l_n}$
Offset term	$-\delta_n$	$-\delta_n$

The OGDBF-w/M algorithm over the BSC can be naturally extended to the D-OGDBF-w/M algorithm over the BSC by considering decoder diversity. The FF of the t -th decoder for the BSC is defined as

$$\check{\Delta}_n^{(\ell)} = \alpha_{n,t} (x_n^{(\ell)} \oplus r_n) + \beta_{n,t} \sum_{m \in \mathcal{M}(n)} \check{s}_m^{(\ell)} - \gamma_{l_n,t} - \delta_{n,t}. \quad (13)$$

V. NUMERICAL RESULTS

In this section, we present extensive numerical results to validate the effectiveness of the proposed algorithms. We first describe our simulation environments and parameter settings. We then compare the proposed algorithms with conventional algorithms in terms of decoding performance. In order to further confirm that the proposed algorithms are beneficial, we discuss their convergence performance and computational complexity.

A. SIMULATION ENVIRONMENTS

To demonstrate the decoding performance of the proposed OGDBF-w/M and D-OGDBF-w/M algorithms, we consider various code configurations with different lengths, rates, and structures:

- 802.11n LDPC codes: We evaluate the (1944, 972) code. The (648, 324) code of the same rate ($R = 1/2$) is included for a shorter code length, while the (1944, 1620) code of the same length is considered for a higher code rate ($R = 5/6$).
- BCH codes: We also include the (63, 45), (127, 106), and (255, 215) BCH codes to evaluate performance in scenarios with a short code length and a highly dense PCM.

The column and row degree distributions of the 802.11n LDPC codes [20] are summarized in Table 2. For the BCH codes, the PCMs are presented in [22], and their degree distributions are provided in Table 3.

TABLE 2. Degree distributions of the considered 802.11n LDPC codes.

802.11n LDPC code	$\Lambda(z)$	$\Gamma(z)$
(1944, 972)	$891z^2 + 729z^3 + 81z^4 + 243z^{11}$	$810z^7 + 162z^8$
(648, 324)	$297z^2 + 270z^3 + 81z^{12}$	$216z^7 + 108z^8$
(1944, 1620)	$243z^2 + 891z^3 + 810z^4$	$81z^{19} + 243z^{20}$

TABLE 3. Degree distributions of the considered BCH codes.

BCH code	$\Lambda(z)$	$\Gamma(z)$
(63, 45)	$2z + 6z^2 + 2z^3 + 4z^4 + 7z^5 + 5z^6 + 9z^7 + 6z^8 + 6z^9 + 10z^{10} + 6z^{11}$	$18z^{24}$
(127, 106)	$4z + 2z^2 + 4z^3 + 2z^4 + 8z^5 + 8z^6 + 13z^7 + 31z^8 + 26z^9 + 14z^{10} + 5z^{11} + 6z^{12} + 4z^{13}$	$21z^{48}$
(255, 215)	$7z + 7z^2 + 3z^3 + 2z^4 + 3z^5 + 5z^6 + 4z^7 + 4z^8 + 2z^9 + 3z^{10} + 6z^{11} + 2z^{12} + 2z^{13} + 11z^{14} + 14z^{15} + 11z^{16} + 12z^{17} + 12z^{18} + 20z^{19} + 12z^{20} + 15z^{21} + 34z^{22} + 29z^{23} + 16z^{24} + 15z^{25} + 4z^{26}$	$40z^{112}$

The proposed algorithms are compared with conventional decoding algorithms including GDBF-w/M, gAD-GDBF, and SP algorithms for decoding performance evaluation. Unlike BF-based algorithms, the SP algorithm deals with floating-point messages and employs flooding scheduling for message updates. As reported in [17], the gAD-GDBF algorithm is compared only for BCH codes. The maximum number of iterations, $\ell_{\max}$, for each algorithm is detailed

TABLE 4. Maximum number of iterations for each algorithm.

Algorithm	$\ell_{\max}$
SP	5, 20* for the considered 802.11n LDPC codes 300 for all the considered BCH codes
GDBF-w/M	300 for all the considered codes
gAD-GDBF [17]	300 for the (63, 45) and (127, 106) BCH codes 800 for the (255, 215) BCH code
OGDBF-w/M and D-OGDBF-w/M	200 for all the considered codes

* $\ell_{\max} = 20$ is applied only to the AWGNC.

in Table 4. Note that it is set to 200 for the proposed algorithms, which is lower than 300 or 800 for the GDBF-w/M and gAD-GDBF algorithms.

The number of decoders in the D-OGDBF-w/M algorithm is set to four for all the considered codes ($T = 4$), where the maximum number of iterations for each decoder is 50, i.e., $\ell_{\max,t} = 50$ for all $t \in [T]$. In contrast, in the gAD-GDBF algorithm [17], six decoders are assigned to both the (63, 45) BCH code and the (127, 106) BCH code, while sixteen decoders are employed for the (255, 215) BCH code. The maximum number of iterations for each decoder in the gAD-GDBF algorithm is 50, as in the D-OGDBF-w/M algorithm.

B. PARAMETER SETTINGS

The decoding parameters for the GDBF-w/M algorithm, namely α , β , γ , and ϵ are specifically configured for each code and are given in Table 5. In particular, the parameters for the (63, 45) BCH code over the AWGNC differ from those in [17], since we have optimized them to ensure a more competitive performance baseline. The parameters and performance results for the gAD-GDBF algorithm are adopted from [17] to establish a conservative and challenging benchmark for comparison.

TABLE 5. Decoding parameters for the GDBF-w/M algorithm.

Code	Channel	α, β, γ , and ϵ
802.11n LDPC code	(1944, 972)	AWGNC 0.75, 1.0, (2, 1, 1, 1), 0.5
		BSC* 1, 2, (4, 2, 2, 1), 0
	(648, 324)	AWGNC 1.0, 1.0, (2, 1, 1, 1), 0.5
		(1944, 1620) 2.0, 1.0, (4, 1, 1, 1), 0.5
BCH code	(63, 45)	AWGNC 7.5, 1.0, (5, 4, 4, 2), 1.0
	(127, 106)	7.2, 1.0, (5, 3, 3, 3), 1.0
	(255, 215)	8.2, 1.0, (5, 3, 3, 3), 2.0

* Parameters are set to integers so that they reflect the discreteness of the BSC.

In order to verify the effectiveness of the proposed offset scheme, the parameters of the proposed algorithms are shared as

- $\alpha_n = \alpha_{n,t} = \alpha$ for all n and t ;
- $\beta_n = \beta_{n,t} = \beta$ for all n and t ; and
- $\gamma_t = \gamma$ for all t .

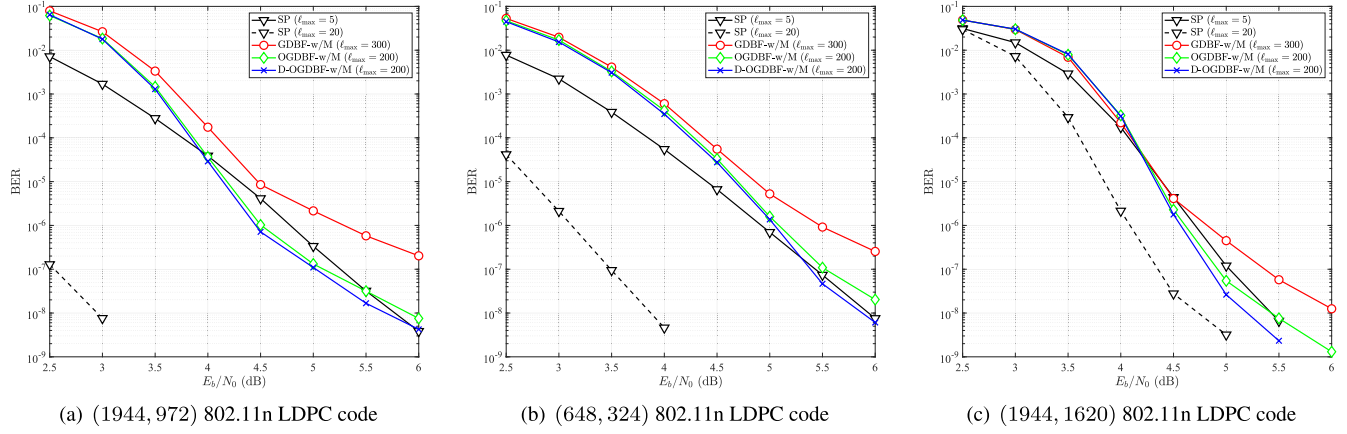


FIGURE 3. BER performances of 802.11n LDPC codes over the AWGNC.

For a fair comparison, the parameters α , β , γ , and ϵ are chosen to be the same as those of the GDBF-w/M algorithm for each code. This means that α , β , and γ are fixed regardless of ℓ_{th} or t . By fixing these baseline parameters rather than performing a joint optimization, we isolate the influence of δ_n and simplify the dimensionality of a parameter search.

Based on the discussions in Sections III and IV, we choose an offset to be shared with all the VNs of the same degree for simplicity and set it to be monotonically increasing with column degree $d(n)$. More specifically, we consider the parameter sharing and monotonicity constraint as

- Parameter sharing
 - OGDBF-w/M: $\delta_{n_1} = \delta_{n_2}$ if $d(n_1) = d(n_2)$;
 - D-OGDBF-w/M: For Decoder t , $\delta_{n_1,t} = \delta_{n_2,t}$ if $d(n_1) = d(n_2)$;
- Monotonicity constraint
 - OGDBF-w/M: $\delta_{n_1} \leq \delta_{n_2}$ if $d(n_1) \leq d(n_2)$;
 - D-OGDBF-w/M: For Decoder t , $\delta_{n_1,t} \leq \delta_{n_2,t}$ if $d(n_1) \leq d(n_2)$.

For highly column-irregular codes such as the (255, 215) BCH code with 26 distinct values of $d(n)$, we employ a grouping strategy assigning the same offset to a group of adjacent values of $d(n)$, thereby preventing a combinatorial explosion.

The chosen offsets in the OGDBF-w/M algorithm for the considered codes are listed in Appendix A, together with ℓ_{th} . As an example, for the (1944, 972) 802.11n LDPC code over the AWGNC, the offset δ_n of VN n in the OGDBF algorithm is selected as 0 for $d(n) = 2$; 1 for $d(n) \in \{3, 4\}$; and 2 for $d(n) = 11$.

The offsets in the D-OGDBF-w/M algorithm, $\delta_{n,t}$, are optimized sequentially for each decoder. The offsets for Decoder 1 are set to zero, i.e., $\delta_{n,1} = 0$ for any n , regardless of $d(n)$. This is because applying an offset term to an unconverged state at early iterations may have a negative impact on decoding performance. For Decoder t ($2 \leq t \leq T$), the offsets are sequentially determined via a greedy search. The resulting offsets are detailed in Appendix B.

The incremental memory overhead for the proposed algorithms relative to the GDBF-w/M algorithm is primarily determined by the storage for the offset values under the given parameter settings. The OGDBF-w/M algorithm uses one offset value per each distinct value of $d(n)$ and thus requires storing at most D values, where D denotes the number of distinct values of $d(n)$. For the D-OGDBF-w/M algorithm, the total number of offset values is up to $D(T - 1)$ via the zero-offset setting of Decoder 1. The overhead for storing the offsets in the D-OGDBF-w/M algorithm is practically marginal within our simulation scope and becomes less significant for long codes, where $D(T - 1) \ll N$ is satisfied. In contrast, for the considered BCH codes, the gAD-GDBF algorithm [17] entails substantially more memory for parameter storage. The $d(n)$ -dependent parameters $\alpha_{n,t}$ and $\beta_{n,t}$ starting from $t = 1$ are employed, along with the decoder-specific parameters such as γ_t for Decoder t . The storage burden of the gAD-GDBF algorithm is further increased due to the use of 6 or 16 decoders, compared to the D-OGDBF-w/M algorithm with only four decoders.

C. ERROR RATE PERFORMANCE

Fig. 3 presents the bit error rate (BER) performance of various 802.11n LDPC codes over the AWGNC. Simulation results demonstrate that across all the considered codes, the OGDBF-w/M algorithm with $\ell_{max} = 200$ consistently exhibits better decoding performance than the GDBF-w/M algorithm with $\ell_{max} = 300$. Further gains are achieved by the D-OGDBF-w/M algorithm at high values of E_b/N_0 through decoder diversity, which reduces the gap to the SP algorithm ($\ell_{max} = 5$) that represents each message as a floating-point value. In addition, the SP algorithm with $\ell_{max} = 20$ is included as a more competitive baseline. For the (1944, 972) code in Fig. 3(a), the D-OGDBF-w/M algorithm shows an approximately 2.5 dB performance loss relative to the SP algorithm ($\ell_{max} = 20$) at $BER = 10^{-7}$. This gap is reasonable as the (1944, 972) code is sufficiently long, and its PCM is sparse enough to enable the near-optimal performance of the SP algorithm, whereas the 1-bit message

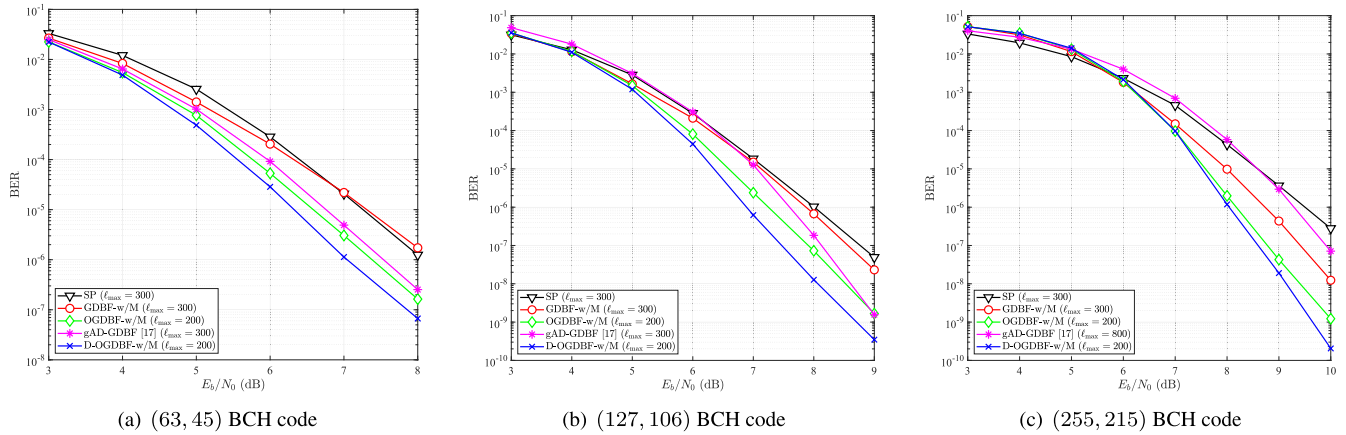


FIGURE 4. BER performances of BCH codes over the AWGNC.

nature of BF-based decoders inherently limits their decoding capability.

The performance gap to the SP algorithm ($\ell_{\max} = 20$) narrows significantly as the code length decreases or the code rate increases. Specifically, for the (648, 324) code, Fig. 3(b) exhibits an approximately 1.8 dB performance gap between the D-OGDBF-w/M and SP algorithms at $\text{BER} = 10^{-7}$. The advantage of the proposed algorithms appears prominent in the (1944, 1620) code, as illustrated in Fig. 3(c). Increasing the code rate leads to a less sparse PCM, which typically degrades the optimality of the SP algorithm. Interestingly, the (1944, 1620) code has better performance than the (1944, 972) code under the conventional GDBF-w/M algorithm, a phenomenon attributable to the lower column-irregularity of the former ($d(n) \in \{2, 3, 4\}$) compared to the latter ($d(n) \in \{2, 3, 4, 11\}$). While this reduced irregularity aligns the performance of the conventional GDBF-w/M algorithm more closely with that of the SP algorithm ($\ell_{\max} = 20$), its impact is further mitigated by the proposed algorithms. Consequently, the D-OGDBF-w/M algorithm reduces its performance gap relative to the SP algorithm ($\ell_{\max} = 20$) to only 0.5 dB at $\text{BER} = 10^{-7}$.

Fig. 4 shows the BER performance of the considered BCH codes over the AWGNC. The simulation results for the (63, 45) BCH code are specifically illustrated in Fig. 4(a). Due to the high density of its PCM, the SP algorithm with $\ell_{\max} = 300$ reveals poor decoding performance. Although the SP algorithm has higher computational complexity, its error-correction capability is comparable to or even inferior to that of the GDBF-w/M algorithm with $\ell_{\max} = 300$. The proposed OGDBF-w/M algorithm with $\ell_{\max} = 200$, in contrast, outperforms the GDBF-w/M algorithm. In particular, the D-OGDBF-w/M algorithm with $\ell_{\max} = 200$ surpasses the gAD-GDBF algorithm [17] with $\ell_{\max} = 300$, where the latter has higher complexity than the former.

The (127, 106) BCH code has higher column-irregularity than the (63, 45) BCH code, as shown in Table 3. This makes an interesting trend in decoding performance. Fig. 4(b) shows that the proposed algorithms handle such irregularity

more effectively in the sense that they have much better performance than the GDBF-w/M algorithm. Specifically, the performance of the OGDBF-w/M algorithm becomes comparable to that of the gAD-GDBF algorithm [17], while the D-OGDBF-w/M algorithm significantly improves the gAD-GDBF algorithm.

The trend becomes even more pronounced for the (255, 215) BCH code with even higher column-irregularity, as depicted in Fig. 4(c). The gAD-GDBF algorithm with $\ell_{\max} = 800$ performs worse than the GDBF-w/M algorithm with $\ell_{\max} = 300$, which suggests that the gAD-GDBF algorithm is not suitable for codes with high column-irregularity. Meanwhile, our proposed algorithms, with only $\ell_{\max} = 200$, achieve a much lower BER than conventional algorithms. The effectiveness of the proposed algorithms is consistently maintained across all the tested BCH codes, confirming the robustness of our offset-based approach.

The proposed OGDBF-w/M and D-OGDBF-w/M algorithms perform well over the BSC. Fig. 5 shows the BER performance of the (1944, 972) 802.11n LDPC code over the BSC. Clearly, the proposed algorithms outperform the existing GDBF-w/M algorithm even though a smaller value of $\ell_{\max}$ is used.

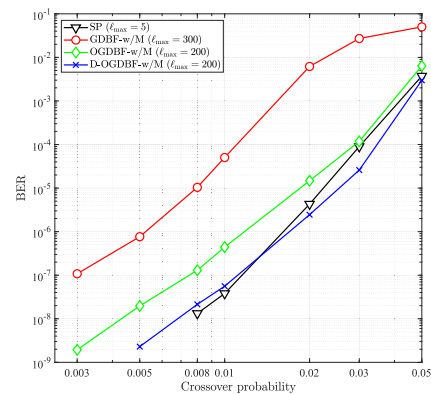


FIGURE 5. BER performance of the (1944, 972) 802.11n LDPC code over the BSC.

D. CONVERGENCE PERFORMANCE

Fig. 6 illustrates the convergence performance of the (1944, 972) 802.11n LDPC code over the AWGNC with respect to the maximum number of iterations, when E_b/N_0 is set to 6 dB. The BER of GDBF-w/M algorithm converges to 2×10^{-7} when the maximum number of iterations is approximately 50, after which it improves only marginally. In contrast, the proposed OGDBF-w/M and D-OGDBF-w/M algorithms decrease the BER to 7.5×10^{-9} and 4×10^{-9} by leveraging the offset term, respectively.

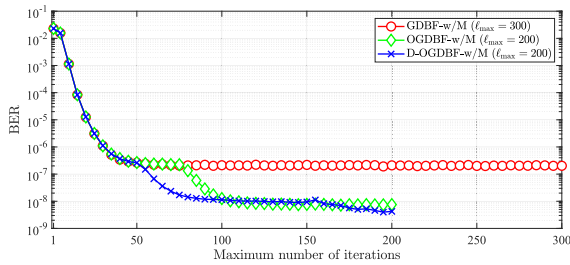


FIGURE 6. Convergence performance of the (1944, 972) 802.11n LDPC code over the AWGNC when E_b/N_0 is 6 dB.

As a result of our parameter selection, the proposed algorithms operate identically to the GDBF-w/M algorithm before $\ell_{th} = 75$ and $\ell_{begin,2} = 50$, respectively. Thereafter, aided by the offset term, the proposed OGDBF-w/M and D-OGDBF-w/M algorithms exhibit a superior error-correction capability compared to the GDBF-w/M algorithm. The improvement is particularly prominent for the D-OGDBF-w/M algorithm, owing to decoder diversity.

E. COMPLEXITY ANALYSIS

The overall complexities of the proposed algorithms are analyzed in terms of the average number of iterations (ANI) and the increase in complexity per iteration. This analysis is performed for the (1944, 972) 802.11n LDPC code over the AWGNC.

The ANIs of the proposed algorithms are presented in Fig. 7. They are significantly lower than the ANI of the GDBF-w/M algorithm at low values of E_b/N_0 , which is mainly due to the decrease of ℓ_{max} from 300 to 200. As E_b/N_0 increases, the ANIs of all the considered algorithms converge to the same value.

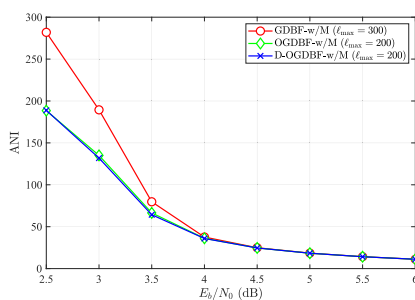


FIGURE 7. ANI of the (1944, 972) 802.11n LDPC code over the AWGNC.

The increase in computational complexity per iteration is also investigated. Compared to the GDBF-w/M algorithm, the proposed algorithms include an offset term within the FFs. Specifically, the FF for the GDBF-w/M algorithm consists of three terms as shown in (2), and summing these terms for N VNs requires $2N$ real additions per iteration. Updates to these individual terms follow a logic common to all such algorithms under our settings, the inclusion of which ensures that the total complexity per iteration exceeds that of $2N$ real additions. For the OGDBF-w/M algorithm, computing the FF in (8) involves N additional real subtractions per iteration only when $\ell \geq \ell_{th}$, where the offset values are fixed as constants. Calculating the FF of the D-OGDBF-w/M algorithm via (9) necessitates N additional real subtractions per iteration solely when $\ell \geq \ell_{begin,2}$ because the offsets for Decoder 1 are set to zero under our parameter selection. As shown in Table 6, the percentage of the frames requiring such additional operations is negligible at high values of E_b/N_0 .

TABLE 6. Percentage of the frames requiring additional offset subtractions compared to the GDBF-w/M algorithm.

E_b/N_0 (dB)	Percentage (%)	
	OGDBF-w/M	D-OGDBF-w/M
2.5	98.072	99.661
3.0	74.530	92.055
3.5	24.138	49.389
4.0	2.812	9.720
4.5	0.272	0.952
5.0	0.054	0.111
5.5	0.015	0.021
6.0	0.005	0.006

The overhead required for managing parameters remains marginal compared to N real subtractions. For the OGDBF-w/M algorithm, only a single comparison per iteration determines whether the inequality $\ell \geq \ell_{th}$ is satisfied. A match between ℓ and $\ell_{begin,t}$ at each iteration acts as the condition for decoder switching in the D-OGDBF-w/M algorithm. The offset values are changed only when the condition is met, as they vary with the decoder index in our settings.

Based on the above discussions, we conclude that the overall complexities of the proposed algorithms are similar to the complexity of the GDBF-w/M algorithm on average. They depend on E_b/N_0 and are summarized as follows.

- Low values of E_b/N_0 : The proposed algorithms have substantially lower ANIs than the GDBF-w/M algorithm. Moreover, the computational overheads, i.e., N real subtractions, are smaller than the complexity per iteration in the GDBF-w/M algorithm.
- High values of E_b/N_0 : All the considered algorithms have similar ANIs, and the offset-related operations are rarely incurred.

VI. CONCLUSION

We presented the OGDBF-w/M algorithm to enhance the decoding performance of irregular parity-check codes, especially column-irregular codes. It introduces an offset term to the FF and effectively adjusts the mean of the FF. This enhances the accuracy of reliability measurements and significantly improves decoding performance. In addition, the proposed D-OGDBF-w/M algorithm further improves decoding performance by leveraging decoder diversity through parameter variation. Numerical results demonstrate that the proposed algorithms achieve significant improvements in decoding performance for irregular parity-check codes, compared to existing algorithms.

APPENDIX A

PARAMETERS FOR THE OGDBF-w/M ALGORITHM

The parameters ℓ_{th} and δ_n for the OGDBF-w/M algorithm are selected as follows.

- (1944, 972) 802.11n LDPC code over the AWGNC:

$$\ell_{th} = 75, \delta_n = \begin{cases} 0, & \text{if } d(n) = 2 \\ 1, & \text{if } d(n) \in \{3, 4\} \\ 2, & \text{if } d(n) = 11 \end{cases}$$

- (1944, 972) 802.11n LDPC code over the BSC:

$$\ell_{th} = 100, \delta_n = \begin{cases} 0, & \text{if } d(n) \in \{2, 3, 4\} \\ 4, & \text{if } d(n) = 11 \end{cases}$$

- (648, 324) 802.11n LDPC code over the AWGNC:

$$\ell_{th} = 100, \delta_n = \begin{cases} 0, & \text{if } d(n) = 2 \\ 1, & \text{if } d(n) = 3 \\ 2, & \text{if } d(n) = 12 \end{cases}$$

- (1944, 1620) 802.11n LDPC code over the AWGNC:

$$\ell_{th} = 100, \delta_n = \begin{cases} 0, & \text{if } d(n) = 2 \\ 1, & \text{if } d(n) \in \{3, 4\} \end{cases}$$

- (63, 45) BCH code over the AWGNC:

$$\ell_{th} = 75, \delta_n = \begin{cases} 0, & \text{if } d(n) = 1 \\ 2, & \text{if } d(n) = 2 \\ 3, & \text{if } d(n) \in \{3, 4\} \\ 4, & \text{if } d(n) \in \{5, 6, 7\} \\ 5, & \text{if } d(n) = 8 \\ 6, & \text{if } d(n) = 9 \\ 7, & \text{if } d(n) = 10 \\ 8, & \text{if } d(n) = 11 \end{cases}$$

- (127, 106) BCH code over the AWGNC:

$$\ell_{th} = 50, \delta_n = d(n) \text{ for all } n$$

- (255, 215) BCH code over the AWGNC:

$$\ell_{th} = 50, \delta_n = \begin{cases} 2, & \text{if } d(n) \in \{1, 2, \dots, 7\} \\ 5, & \text{if } d(n) \in \{8, 9, \dots, 14\} \\ 7, & \text{if } d(n) \in \{15, 16, \dots, 21\} \\ 10, & \text{if } d(n) \in \{22, 23, \dots, 26\} \end{cases}$$

APPENDIX B

OFFSETS FOR THE D-OGDBF-w/M ALGORITHM

The offsets $\delta_{n,t}$ ($2 \leq t \leq 4$) for the D-OGDBF-w/M algorithm are listed in Tables 7, 8, 9, 10, and 11.

TABLE 7. Offsets $\delta_{n,t}$ for the (1944, 972) 802.11n LDPC code over the AWGNC and BSC.

AWGNC				BSC			
$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4
2	0	0	0	2	0	0	0
3	1	1	0	3	0	0	0
4	1	1	2	4	0	2	0
11	3	2	2	11	4	5	1

TABLE 8. Offsets $\delta_{n,t}$ for the (648, 324) and (1944, 1620) 802.11n LDPC codes over the AWGNC.

(648, 324)				(1944, 1620)			
$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4
2	0	0	0	2	0	0	0
3	1	0	1	3	0	1	1
12	2	2	3	4	1	1	2

TABLE 9. Offsets $\delta_{n,t}$ for the (63, 45) BCH code over the AWGNC.

$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4
1	0	0	0	5	4	2	0	9	6	4	4
2	2	0	0	6	4	3	0	10	7	4	6
3	4	0	0	7	5	3	1	11	8	5	7
4	4	1	0	8	6	4	2				

TABLE 10. Offsets $\delta_{n,t}$ for the (127, 106) BCH code over the AWGNC.

$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4
1	1	3	5	6	6	6	5	11	11	9	10
2	2	3	5	7	7	6	10	12	12	9	10
3	3	3	5	8	8	6	10	13	13	9	10
4	4	3	5	9	9	9	10				
5	5	6	5	10	10	9	10				

TABLE 11. Offsets $\delta_{n,t}$ for the (255, 215) BCH code over the AWGNC.

$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4	$d(n) \backslash t$	2	3	4
1	2	1	3	10	5	6	3	19	7	10	10
2	2	1	3	11	5	6	3	20	7	10	10
3	2	1	3	12	5	6	3	21	7	10	10
4	2	1	3	13	5	6	3	22	10	10	10
5	2	1	3	14	5	6	10	23	10	10	10
6	2	1	3	15	7	6	10	24	10	10	10
7	2	1	3	16	7	6	10	25	10	10	10
8	5	1	3	17	7	6	10	26	10	10	10
9	5	1	3	18	7	6	10				

REFERENCES

- [1] R. G. Gallager, "Low-density parity-check codes," *IRE Trans. Inf. Theory*, vol. 8, no. 1, pp. 21–28, Jan. 1962.
- [2] T. J. Richardson, M. A. Shokrollahi, and R. L. Urbanke, "Design of capacity-approaching irregular low-density parity-check codes," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 619–637, Feb. 2001.
- [3] T. J. Richardson and R. L. Urbanke, "The capacity of low-density parity-check codes under message-passing decoding," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 599–618, Feb. 2001.
- [4] J. Hou, P. H. Siegel, and L. B. Milstein, "Performance analysis and code optimization of low density parity-check codes on Rayleigh fading channels," *IEEE J. Sel. Areas Commun.*, vol. 19, no. 5, pp. 924–934, May 2001.
- [5] X. Zhang and P. H. Siegel, "Quantized iterative message passing decoders with low error floor for LDPC codes," *IEEE Trans. Commun.*, vol. 62, no. 1, pp. 1–14, Jan. 2014.
- [6] S. K. Planjery, D. Declercq, L. Danjean, and B. Vasic, "Finite alphabet iterative decoders—Part I: Decoding beyond belief propagation on the binary symmetric channel," *IEEE Trans. Commun.*, vol. 61, no. 10, pp. 4033–4045, Oct. 2013.
- [7] T. Wadayama, K. Nakamura, M. Yagita, Y. Funahashi, S. Usami, and I. Takumi, "Gradient descent bit flipping algorithms for decoding LDPC codes," *IEEE Trans. Commun.*, vol. 58, no. 6, pp. 1610–1614, Jun. 2010.
- [8] O. A. Rasheed, P. Ivaniš, and B. Vasic, "Fault-tolerant probabilistic gradient-descent bit flipping decoder," *IEEE Commun. Lett.*, vol. 18, no. 9, pp. 1487–1490, Sep. 2014.
- [9] G. Sundararajan, C. Winstead, and E. Boutillon, "Noisy gradient descent bit-flip decoding for LDPC codes," *IEEE Trans. Commun.*, vol. 62, no. 10, pp. 3385–3400, Oct. 2014.
- [10] H. Cui, J. Lin, S. Song, and Z. Wang, "A new probabilistic gradient descent bit flipping decoder for LDPC codes," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Sapporo, Japan, May 2019, pp. 1–5.
- [11] H. Cui, J. Lin, and Z. Wang, "An improved gradient descent bit-flipping decoder for LDPC codes," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 66, no. 8, pp. 3188–3200, Aug. 2019.
- [12] V. Savin, "Gradient descent bit-flipping decoding with momentum," in *Proc. 11th Int. Symp. Topics Coding (ISTC)*, Montreal, QC, Canada, Aug. 2021, pp. 1–5.
- [13] W. Min and K. Yang, "Channel-aware gradient descent bit-flipping algorithm for LDPC codes," *IEEE Commun. Lett.*, vol. 29, no. 4, pp. 809–813, Apr. 2025.
- [14] J. Milojković, S. Brkić, P. Ivaniš, and B. Vasić, "Generalized adaptive diversity gradient descent bit-flipping with a finite state machine," *Entropy*, vol. 27, no. 1, p. 49, Jan. 2025.
- [15] S. Brkić, P. Ivaniš, and B. Vasić, "Adaptive gradient descent bit-flipping diversity decoding," *IEEE Commun. Lett.*, vol. 26, no. 10, pp. 2257–2261, Oct. 2022.
- [16] C. He, K. Deng, S. Song, and Z. Wang, "Column-weighted probabilistic GDBF decoder for irregular LDPC codes," in *Proc. IEEE Comput. Soc. Annu. Symp. VLSI (ISVLSI)*, vol. 47, Foz do Iguacu, Brazil, Jun. 2023, pp. 1–6.
- [17] J. Milojković, S. Brkić, P. Ivaniš, and B. Vasić, "Learning to decode linear block codes using adaptive gradient-descent bit-flipping," in *Proc. 12th Int. Symp. Topics Coding (ISTC)*, Brest, France, Sep. 2023, pp. 1–5.
- [18] M. G. Luby, M. Mitzenmacher, M. A. Shokrollahi, and D. A. Spielman, "Improved low-density parity-check codes using irregular graphs," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 585–598, Feb. 2001.
- [19] S. Myung, K. Yang, and J. Kim, "Quasi-cyclic LDPC codes for fast encoding," *IEEE Trans. Inf. Theory*, vol. 51, no. 8, pp. 2894–2901, Aug. 2005.
- [20] *IEEE Standard for Information Technology—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 5: Enhancements for Higher Throughput*, Standard 802.11n-2009, Oct. 2009.
- [21] S. Lin and D. J. Costello, *Error Control Coding*, 2nd ed., Upper Saddle River, NJ, USA: Pearson, 2004.
- [22] *Generalized AD-GDBF Decoder Database*. Accessed: Apr. 1, 2026. [Online]. Available: <https://github.com/milojko94/gAD-GDBF>
- [23] W. Ryan and S. Lin, *Channel Codes: Classical and Modern*. Cambridge, U.K.: Cambridge Univ. Press, 2015.



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